Eager for adventure, Kolob drops out of the starship's belly and plunges downward, blasting his rockets gently at first, then harder, to resist the swirl of space and remain directly below the ship. At first Kolob has no problems. But when he reaches a circumference of 833 kilometers, 56 percent larger than the horizon, his laser light brings the message, "I can't resist the swirl; 1 can't; 1 can't!" and like a rock caught up in a tornado he gets dragged into a circulating orbit around the hole.[41]

"Don't worry," you reply. "Just do your best to resist the swirl, and continue to descend until you are ten meters above the horizon."

Kolob complies. As he descends, he is dragged into more and more rapid circulating motion. Finally, when he stops his descent and hovers ten meters above the horizon, he is encircling the hole in near perfect lockstep with the horizon itself, 270 circuits per second. No matter how hard he blasts to oppose this motion, he cannot. The swirl of space won't let him stop.

"Blast in the other direction," you order. "If you can't circle more slowly than 270 circuits per second, try circling faster."

Kolob tries. He blasts, keeping himself always 10 meters above the horizon but trying to encircle it faster than before. Although he feels the usual acceleration from his blast, you see his motion change hardly at all. He still circles the hole 270 times per second. And then, before you can transmit further instructions, his fuel gives out; he begins to plummet downward; his laser light zooms through the electromagnetic spectrum from green to red to infrared to radio waves, and then turns black with no change in his circulating motion. He is gone, down the hole, plunging toward the violent singularity that you will never see.

After three weeks of mourning, experiments, and telescopic studies, your crew begin to build for the future. Bringing in materials from distant planets, they construct a girder-work ring around the hole. The ring has a circumference of 5 million kilometers, a thickness of 552 kilometers, and a width of 4000 kilometers. It rotates at just the right rate, two rotations per hour, for centrifugal forces to counterbalance the hole's gravitational pull at the ring's central layer, 276 kilometers from its inner and outer faces. Its dimensions are carefully chosen so that those people who prefer to live in 1